

A1 kernel identity – zero-momentum mass vs shell mass (complete-the-square)

TECT verification-first repository · claim A1-KERNEL-IDENTITY

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1. Purpose and scope

This card removes a naming conflation in A1-KERNEL-CONV: the *zero-momentum mass* $r := K(0)$ and the *shell mass* $m_{\text{sh}}^2 := K(q_*)$ were both written μ^2 , yet they differ by Yq_*^4 . This is the algebraic identity; the analytic hypothesis it feeds is A1-SCALAR-ANALYTIC-BRANCH, and the parameter-consistency gate is A1-PRODUCTION-KERNEL-MANIFEST. Verification: [codes/foundations/a1_kernel_checks.py](#) (a1_kernel_checks.py, all PASS).

2. The identity

Theorem (A1-KERNEL-IDENTITY). *For $Y > 0$, $Z < 0$ and any $r \in \mathbb{R}$, the kernel $K(q) = r + Z|q|^2 + Y|q|^4$ satisfies*

$$q_*^2 = -\frac{Z}{2Y} > 0, \quad m_{\text{sh}}^2 := K(q_*) = r - \frac{Z^2}{4Y}, \quad K(q) = m_{\text{sh}}^2 + Y(|q|^2 - q_*^2)^2, \quad (1)$$

and

$$r = K(0) = m_{\text{sh}}^2 + Yq_*^4. \quad (2)$$

The completion of the square is unconditional algebra (residual 5.7×10^{-14} over a random sweep, assert `kernel_identity_complete_square`); $Z < 0$ makes $q_*^2 > 0$ a physical shell radius. Thus r (zero momentum) and m_{sh}^2 (shell) coincide only when $q_* = 0$ ($Z = 0$); in general $r = m_{\text{sh}}^2 + Yq_*^4 > m_{\text{sh}}^2$.

3. Naming (binding for A1)

The symbol μ^2 is deprecated for A1; use m_{sh}^2 (`mu2_shell`) for the shell mass $K(q_*)$ and r (`r_zero/r_k0`) for the zero-momentum mass $K(0)$. On a fixed periodic cell, $\lambda_0(L) := \min_{k \in \Lambda^*} K(k) \geq m_{\text{sh}}^2$, with equality only if a lattice mode lies exactly on the shell $|k| = q_*$.

4. Devil’s-advocate

(α) “*The identity is trivial.*” VALID — it is elementary, but the conflation it removes is not: the A2/A3 analytic hypothesis is $m_{\text{sh}}^2 > 0$, while the solver template’s `mu2` is r (§A1-PRODUCTION-KERNEL-MANIFEST). (β) “ *$Z < 0$ is an extra assumption.*” DISMISSED — $Z < 0$ is exactly what makes the Brazovskii shell $q_*^2 > 0$ exist; with $Z \geq 0$ the minimum is at $q = 0$ and there is no shell. (γ) “ *$Y > 0$ needed?*” Yes — $Y > 0$ for the completion and for the shell to be a minimum (not a maximum). Quantitative sanity check: residual 5.7×10^{-14} ; $r - (m_{\text{sh}}^2 + Yq_*^4) = 0$ identically.

5. Result footer

Result ID: A1-KERNEL-IDENTITY
Precise statement: For $Y>0$, $Z<0$: $K=r+Z q^2+Y q^4 = m_{sh2}+Y(q^2-q^2)^2$,
 $q^2=-Z/2Y$, $m_{sh2}=r-Z^2/4Y$, $r=K(0)=m_{sh2}+Y q^4$.
 r (zero-momentum) and m_{sh2} (shell) are distinct.
Scope: algebraic identity; $Y>0$, $Z<0$.
Dependencies: A1-KERNEL-CONV (the kernel form).
Evidence grade: PROVED (analytic) + EXECUTED (a1_kernel_checks.py; this card's
assert: kernel_identity_complete_square).
Reproduction command: python codes/foundations/a1_kernel_checks.py
Expected output: all PASS (this card: kernel_identity_complete_square); exit 0.
Tier: T6 PROVED (submitted for operator sign-off).
No-overclaim statement: Naming/identity only; the positivity branch is
A1-SCALAR-ANALYTIC-BRANCH; production consistency is
A1-PRODUCTION-KERNEL-MANIFEST.
Next required action: Operator sign-off; re-point A2/A3 to the analytic branch.